

Statistics & Quantitative Methods for Data Science

Course Overview — Silvadew Institute

This course gives learners a practical, project-driven foundation in statistics and quantitative methods for data science. Over six weeks, learners move from descriptive statistics and probability through inferential methods, regression, and statistical foundations for machine learning — closing with a project-intensive final week that includes peer learning with guided support.

At a Glance

Duration	6 weeks
Delivery	Jupyter notebooks and whiteboard sessions
Format	10 lectures + 5 labs, one lab weekly (Weeks 1–5), 2 major project weeks + peer learning (Week 6)
Major Projects	3 (Trigrams for LLM Foundations, Regression, Classification)
Tools	Python — NumPy, Pandas, Matplotlib

Weekly Structure (Weeks 1–5)

Each week runs two lectures — delivered through a mix of Jupyter notebooks and whiteboard sessions — plus one hands-on lab, building a total of 10 lectures and 5 labs (one weekly) across the first five weeks.

Week 1	Introduction to Statistics for Data Science Types of data, descriptive statistics, data visualization, and Python tooling (NumPy, Pandas, Matplotlib).
Week 2	Probability Fundamentals Probability distributions, sampling techniques, and the Central Limit Theorem.
Week 3	Inferential Statistics Confidence intervals, hypothesis testing, p-values, and statistical significance.
Week 4	Correlation & Regression Simple and multiple regression, exploratory data analysis (EDA), and feature relationships.
Week 5	Statistics for Machine Learning Statistical methods for ML, feature selection, model evaluation metrics, and an introduction to A/B testing.

Week 6 — Capstone Projects & Peer Learning

The final week shifts from lecture/lab pacing to project depth. Learners complete three major projects and work through them in peer-learning groups with instructor guidance, culminating in a real-world data analysis deliverable, statistical reporting and visualization, and a portfolio-ready capstone.

1. Trigrams — Foundations of LLMs

Building and analyzing trigram models to explore the statistical foundations underlying large language models: n-gram probability, sequence prediction, and language statistics.

2. Regression Project

An applied regression project translating Week 4's correlation and regression concepts into a full modeling workflow — from EDA to model fitting and interpretation.

3. Classification Project

A classification project applying statistical methods for ML and model evaluation metrics from Week 5 to a real-world predictive task.

Peer learning groups work through each project together, with guided instructor support throughout — reinforcing statistical reasoning through collaboration and real feedback.